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Sustainable joint forest management through bargaining: a bilateral monopoly gaming approach

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Abstract

The failure of forest owners and managers to exclude user groups from use of the resource forces them to opt for a collaborative management approach, termed as joint management or comanagement. Joint forest management has been compared with agriculture crop sharing arrangements. With the objective of making joint forest management a sustainable venture, a model based on the cooperative/bargaining nature of the agreements for sharing the forest produce between the two partners has been developed. Shares of the two partners have been worked out for two cases of joint forest management from the state of West Bengal in India, and outcomes have been compared with existing sharing arrangements.

Keywords: Bargaining game; Bilateral monopoly; Fixed threat cooperative game; Joint forest management; Developing countries

1. Introduction

Deforestation in developing countries has recently become a major concern (Allen and Barnes, 1985; World Commission on Environment and Development (WCED), 1987, pp. 5, 13 and 151; Repetto, 1988, pp. 3–12; Bundestag, 1990, p. 239). Disagreement on the magnitude, causes and consequences of deforestation is common (Fearnside, 1982; Lugo and Brown, 1982). Nevertheless, with the help of statistical analysis, Allen and Barnes (1985) confirmed that deforestation is significantly related to population growth, agricultural expansion and past rate of wood production in a sample of 28 countries from Africa, Asia and Latin America. It seems that the main cause of deforestation in these areas

is not overfelling for supply to industries but the inability of forest owners to exclude various user groups from the resource which in theory is under a well-defined property regime (Bundestag, 1990, p. 262; Poffenberger, 1990, p. xxi). These user groups cut timber for fuel, lop trees for fodder and graze their cattle in the area so that regeneration is difficult to obtain; and damage wildlife either by hunting or destroying their habitat. This issue has not attracted much attention from economists. Whenever it has, the focus of discussion has been the definition of property rights so that an impression is created that once property rights are properly defined the deforestation problem will automatically be solved (Mendelsohn, 1991; Barbier et al., 1991). We contend that this is not so. India is an example of a country where the rights of the forest owner (the state) are very clearly defined as also are

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those of local people living near it, and yet forest reserves have been used as an open access resource in many cases (Kant et al., 1991, p. 2). There are many other examples, some of them from the industrialized countries.

Mendelsohn (1991) argued for private property rights to well-defined parcels (optimal size) of forest lands through either sale in the financial market or by lottery. However, owing to large population, heavy dependence of a large number of people on forest resources for daily subsistence needs^a, and very small per capita forest area (0.266 ha for tropical Asia (FAO, 1982; p. 26)), the proposition of privatization of forests is not feasible for most countries in the Third World. This is so because private ownership will either be concentrated in the hands of a few large companies or diffused among the population at large. In the former scenario the fate of the forests will be no different than present because local user groups will continue to exert pressure for their daily subsistence needs. The experiences of some countries like Indonesia and Malaysia also indicate that in the given policy environment, large private firms could not manage forests with an objective of long-term sustainable management (Repetto and Gillis, 1988, pp. 26–32). In the latter scenario forests would be divided into very small parcels, say of 0.2 ha or so, and handing over these patches to individuals would lead to a process in which the forest area would lose its meaning as a forest ecosystem. These small patches of few trees cannot be meaningfully managed as forest. So, attention needs to be paid to the sheer impossibility of property owners and managers excluding desperate local populations (the user groups) from using the resource in a manner that not only exceeds the latter's legal rights but is also more than the sustainable capacity of the forest to produce, and leads to destruction of a forest to a great extent. The major conflict between the forest owner and communities is of fixing the priorities for different needs. The forest owner (government) gives the top

priority to national needs, while the communities give the top priority to local needs. The two needs disagree on the issues of the basic product of trees. For example, timber may be desired by the nation as a whole but the local community may want poles, the nation may want pulp but the local community may want small timber, and so on. The consequences are conflicts on choice of species, rotation, etc. These conflicts can be resolved only in a cooperative fashion. Social systems and cultural values in most Third World countries also favour the effective participation of local communities in the decision-making processes of forest management (WCED, 1987, p. 47).

One of the noticeable efforts in the Third World to resolve such conflicts is the Joint Forest Management Programme in the state of West Bengal in India (Malhotra and Poffenberger, 1989), which deals with the rehabilitation of degraded Sal (*Shorea robusta*, Gaertn) forests. Other programmes based on the collaborative approach are the Community Forest Stewardship Programme of the Philippines (Gibbs et al., 1990), community management of forests of Nepal (Arnold and Cambell, 1986; Karki and Dixit, 1991), CAMPFIRE (Community Areas Management Program For Indigenous Resources) of Zimbabwe and ADMAD (Administrative Management Design) for game management areas of Zambia (Forests, Trees and People Newsletter, 1991). Similar programmes like comanagement agreements in Manitoba (Alison and Berkes, 1991), community forestry in British Columbia (Pinkerton, 1991), and the joint stewardship agreement of the Ontario Government with the Bear Island Band of Temagami (Forestry Sectoral Task Force, 1992) have also been reported from the industrialized world.

To be successful, agreements on joint forest management (JFM) should include the 'rights and duties' of both partners, and sharing arrangements for all the products. The success of such a programme will depend upon the fairness of benefit sharing arrangements as perceived by the two parties. In the absence of an equitable sharing arrangement, a programme may certainly get initiated and may even continue for

^aThree-fourths of the population in developing countries depend on fuelwood and other traditional fuels for their daily energy needs (FAO, 1982, p. 24).

some time, but it will not sustain itself. In this paper an attempt has been made to show the broad parameters of a methodology to determine the user group's share such that its long-term interest can be maintained in cooperation with a forest owner in forest management.

To this end, JFM has been compared with agriculture crop sharing arrangements, and arguments have been put forward to show that JFM can be seen as an example of a bilateral monopoly. Literature regarding modelling bilateral monopolies as a bargaining game, and its application for determining shares in crop-sharing arrangements has been reviewed. A general model for sharing the forest produce under JFM has been developed, and different real life situations have been highlighted. Outcomes of the model for one specific situation have been discussed. Finally, we close with a discussion of a case of JFM from India, and some suggestions for long-term cooperation of the user group in that case.

In view of the case study and the relative familiarity of the authors with the Indian forest scene the arguments have been developed with particular reference to India. Nevertheless, the broad outcomes should be applicable to most JFM issues around the world.

2. Joint forest management and system of agriculture crop sharing

In principle, JFM is similar to the share cropping system in agriculture. The variations of sharing arrangements' stipulations is almost the same as the variation in share cropping arrangements — variation in share of community, variation regarding provision of inputs, etc.^b. The main difference is in the gestation period which is only a few months in agriculture while in forestry, it may be anywhere from about 10 to 40 years in the tropics and longer in the temperate

regions. But, even in forestry the user group (community) usually starts getting annual returns after a period of 1 or 2 years. These returns are due to what are often called the non-timber forest products and, in fact, may be critical to the livelihood of individuals in the community.

Share cropping has drawn comments from economists at least since the time of Arthur Young and Adam Smith in the late eighteenth century. The attitude of these early commentators was universally one of disapproval and occasionally of strong condemnation. Since Alfred Marshall's (1920, p. 644) piece on share cropping in which he concluded that 'production under share cropping will not be efficient', relatively little attention has been paid to share cropping by economists (Quibria and Rashid, 1984).

It was only recently that Bardhan and Srinivasan (1971, 1974) developed a fairly complete scheme of analysis in which rental share is regarded by the parties of contract as a 'price like variable'. The 'principal and agent problem' approach was used by Cheung (1969) which was generalized by Newbery (1975) to explain the phenomenon of share cropping. In this approach, both the rental share and minimum labour input per unit of land are jointly determined by the landlord's desire to maximize his income subject to the condition that the tenant's income does not fall below his alternative earnings in a competitive labour market.

The competitive model of Bardhan and Srinivasan cannot be used to analyse JFM cases because, first, lease allotment for JFM cannot be termed as competitive because there is a single land owner (usually the government) and single user group. Sometimes there may be more than one user group but their number is usually very small. No arrangement can be made with a far away community on the same footing as with a nearby community, because it is bound to lead to failure of exclusion of the nearby community from the resource, and so failure of the whole effort.

Secondly, in the case where the owner is the State, the objective of the land owner cannot be taken to be maximization of its income. Forest

^bIn India's JFM programme, community shares in final timber harvest have been fixed differently. The shares are 25% in West Bengal, 30% in Madhya Pradesh, 50% in Maharashtra. Similarly, different arrangements for the provision of inputs have been proposed.

policies based on maximization of revenue at the cost of social welfare have resulted in deforestation (Repetto, 1988, pp. 12–16). A welfare state (as most states in the world today are) is morally bound to have one of its objectives to maximize social welfare. For example, Article 38 of the Indian constitution is quite clear in expecting the State to maximize social welfare (Das, 1978, p. 392). Therefore, it is more reasonable to assume that the objective of the State is to evolve a sustainable resource management system that maximizes the total gain to society as a whole.

Joint forest management is an attempt to forge a partnership based on evolving joint management objectives in which communities share both responsibilities and proceeds (Campbell, 1992). Poffenberger and Singh (1991) state that JFM represents a process of decentralized empowerment and making communities equal partners in forest management. Thus, in this programme, the relationship between state and communities is not that of a 'principal and agent' but of two equal partners. We, therefore, do not consider the Cheung (1969) and Newbery (1975) approach to be suitable for JFM. In fact, we feel that before the JFM programme the state might have been treating itself as 'principal' and the neighbouring communities as 'agents' in India and other countries. The failure of this approach to yield positive results has forced the owners and analysts to look for alternatives.

3. Joint forest management as a bilateral monopoly

In situations where the government owns forest land and enjoys a monopoly in supplying forest products to the community, the latter, also enjoys monopsony in demanding much, if not all, of the products. It is almost a textbook case of a bilateral monopoly. This situation is reinforced by the fact that the community is usually the only supplier of the type of service which is necessary for the management of these forests, and the forest owner the only possible employer. The type of service in question includes the practice of self-

restraint, and keeping a watch on the forest to protect it from outsiders.

We assert that most cases of JFM fall within the definition of a bilateral monopoly. In particular, it must be emphasized that the individual members of a user group do not deal separately with the owner, at least not in the Third World.

To understand the economic character of JFM, a true understanding of community is necessary. As Thurow (1983, pp. 222–223) points out 'Societies are not merely statistical aggregations of individuals engaged in voluntary exchange but something more subtle and complicated. A group or community cannot be understood if the unit of analysis is the individual taken by himself. A society is clearly something greater than the sum of its parts.' Thurow distinguishes 'private personal preference' from 'individual social' ones. Following Thurow, Daly and Cobb (1989, p. 7) proposed that an individual is '*Homo economicus* as person in community' rather than '*Homo economicus* as pure individual'. This character of an individual is perfectly reflected in the observations of many scholars, regarding peasant village life. Maine (1905, p. 262) said 'An Indian village community is an organized society.' James Scott (1976, p. 40) portrayed the village in a precapitalist peasant society as a key institution, characterized by a variety of social arrangements designated to insure village members against a subsistence crisis. Wade (1988, p. 5) showed that villages in south Indian uplands, sustain a public realm of sophistication. Guha (1989, p. 21) remarked that owing to a strong sense of clanship, peasants in the Himalayas often derived their caste name from the village they inhabited. The institutional expression of this solidarity is the village panchayat (council). Numerous recent publications on successful management of common resources by communities (National Research Council, 1986; McCay and Acheson, 1987; Berkes, 1989; Pinkerton, 1989; Bromley, 1992) demonstrate the nature of an individual to be '*Homo economicus* as person in community'. The recent World Bank (1992) Development Report also observes that group action is deeply rooted in many societies for managing land resources.

In such community settings, the interest of all

a community's members is represented by a selected/elected body, i.e. committee. Thus, there are only two players, i.e. the government and the local community, in the game. The role of a committee is similar to the role of a union in the case of the wage bargaining phenomenon (Menil, 1971).

4. Solution of a bilateral monopoly as a bargaining game

Numerous approaches have been suggested to the problems of bilateral monopoly (Marshall, 1920, pp. 493–494; Bowley, 1928; Zeuthen, 1930, pp. 63–100; Hicks, 1935; Tintner, 1939; Stigler, 1946, pp. 266–267; Fellner, 1947; Nash, 1950, 1953; Fouraker, 1957; Schilling, 1957). Variation among solutions stems from variation in the assumptions and conditions which are made regarding the form of the negotiations employed by, or imposed upon, the participants. Siegel and Fouraker (1960, pp. 3–14), identified three forms of negotiations: price leadership, equal bargaining strength and strict all-or-none bids for bilateral monopolies. In the case of JFM, contractual agreements between a government and a community are agreed upon only after the community (usually the less sophisticated party) is made fully aware of all the consequences of such arrangements. Both the parties agree to abide by an agreement with a sense of commitment and complete knowledge of gains from such an arrangement. They are also aware of the cost of breach of trust and non-existence of such a contract. We believe that both partners, i.e. the owner and the community, are in a position of equal bargaining strength.

The challenge is in choosing an appropriate concept for the solution of this game. The non-cooperative game solution concepts are appropriate for situations in which neither party is offering to cooperate nor attempting to induce the other to cooperate (Shubik, 1988, p. 86). In JFM, the situation is entirely different. Both parties are making efforts to obtain cooperation from each other. The challenge reduces to selection of an appropriate cooperative solution concept. The

alternatives available can be Walrasian equilibrium, the Shapley value, the core, Von Neumann and Morgenstern solution set, or Nash solution. Within the Nash program, each of these and other cooperative solution concepts are seen as an attempt to describe the outcome of a non-cooperative negotiation game without formalizing the negotiation procedure. To be useful, a cooperative solution procedure must be applicable to a fairly wide class of negotiation procedures. In some cases, such as the Shapley value, the concept is best seen as an 'average' of what is to be expected over such a class of negotiation procedures. In other cases such as the core or von Neumann and Morgenstern solution set, the concept does not supply a unique outcome (Binmore and Dasgupta, 1987, p. 8). Therefore, for the cooperative situations of JFM, the Nash (1950, 1953) solution of the bargaining game seems to be the best option available. The choice of Nash solution is also supported by the outcomes of negotiation for other common resources, i.e. groundwater disputes in the Raymond Basin, West Basin, Central Basin and San Gabriel Basin in the USA where the solution was close to the Nash solution (Ostrom, 1990, p. 231). The concept of the bargaining game was also used by Bell and Zusman (1976) to explain the different rents for agricultural crop-sharing contracts in specific situations where the authors could identify the existence of non-tradable factors such as managerial skills and draught power, and thus market imperfections.

Equal partnership in forest management bargaining is particularly relevant to those developing countries, such as India, Pakistan and Bangladesh, where the rural population in the vicinity of forests continues to believe that the forests belonging to them were unjustly taken over by the former colonial government (Guha, 1989, p. 57; Agrawal and Narain, 1989, p. 13). In such a historical, cultural and socio-economic environment, it is but natural to accept the fact that both the government and the people have rights and duties for managing and using the forests. Thus, the specific features of JFM, and basic characteristics of the local rural environment, strengthen

our argument for using cooperative game theory for this issue.

5. Two-person bargaining/cooperative game and JFM

In general terms, a bargaining problem is idealized by assuming that the two participants are rational, that each can compare accurately their desires for various goods and services, that they are equal in bargaining skills and that each has a full knowledge of the tastes and preferences of the other (Nash, 1950). We consider a two-person game of exchange in which the strategies are decisions to trade commodities on various specific terms. In this game there are no reprisals possible for either party if their terms are not met. Both have to go home with whatever they brought to the bargaining table. The pair of pay-offs in the event of no trade is status quo and Bacharach (1976, p. 112), has called it the 'fixed threat'. The game is fully specified by a cooperatively attainable set in the pay-off or the utility spaces and the status quo or fixed threat point.

A unique equilibrium solution for a bargaining or two-person fixed threat cooperative game, as proposed by Nash (1950, 1953), is the set of utilities or pay-offs which maximizes the product of the two players' utility gains measured from the status quo /fixed threat point or the product of their pay-off increments. The solution is made on the grounds that it satisfies the four conditions, Pareto-optimality, interpersonal non-comparability, symmetry and independence of irrelevant alternatives (Bacharach, 1976, pp. 94–95). Even though Nash included bilateral monopoly in bargaining games, Kaneko (1981) proposed a different approach for reaching a Nash solution in the case of a bilateral monopoly.

Nash proved that the proposed outcome is an equilibrium outcome for a two-person bargaining game. Kaneko (1981) modified the approach to incorporate the case of a bilateral monopoly, and adopting Selten's (1975) approach of 'perfect equilibrium point', a point to have the stability against small imperfections of rational-

ity on every information set, proved that Nash solution is an equilibrium solution.

This result can be used to find a solution of bargaining situations in JFM. Variations may occur from place to place, but the prebargaining JFM situation can be characterized as follows.

The owner (government) threatens that it will refuse to share the forest land with the local community, and the community threatens that it will resort to free use of forest produce. The owner may have to resort to policing in the face of the community carrying out its threat, and the fixed threat pay-off of the owner is equal to the revenue earned from the sale of forest products and the penalties that offenders are charged minus the cost of management and policing. The penalties imposed by government consist of the value of the product taken away from the forest, and a punishment fee. The punishment is fixed according to the nature of the offence. The economic argument of imposing very high or infinite penalties to control the offence has practical limitations. A high penalty beyond the paying capacity of the offender, cannot be recovered easily. In cases of very high penalties imposed by the government, the offender may either reoffend or may challenge the authority of the government in a court of law. Infinite or very high penalties cannot withstand the law of natural justice. Therefore, the fixed threat pay-off of the owner cannot be very high. The fixed threat pay-off of the community is equal to the value obtained from the illicit extraction of forest produce minus penalties paid to the owner. Let us say that W_o and W_c denote the fixed threat pay-off per hectare per year for the owner and the community, respectively.

For the model in the main body of the paper, some very strong assumptions have been made. It is done to keep the model simple. Some of these assumptions have been relaxed in the Appendix.

We assume that decision making is decentralized, and so the local forest officers can take situation-specific decisions within the broad framework of their state government. Fixed threat pay-offs W_o and W_c are assumed to be constant over time, for the period under consideration. We also suppose the extent of forest area

to be jointly managed by a community, and that the rotation length to be fixed is negotiable between the owner and the community, and so both will look for an area and rotation which give maximum total pay-off gain. Further, we assume that forest area and rotation are denoted by A and R , respectively. Since the owners have attempted JFM in desperation, in decentralized decision making, they may have the same rate of time preference, as the community. Therefore, we assume that discount rate for use by both parties is the same and that its value is such as to have its continuous equivalent as r . Let the per unit area yield of products in the form of sawlogs, pulpwood, etc. (here called timber products) from the forest at rotation be Q_t units, and let this produce be sold in the open market at price P per unit.

We also assume that Q_t is bounded and

$$Q_t = F(R, A) \quad \partial F / \partial R > 0, \partial^2 F / \partial R^2 < 0$$

and $\partial F / \partial A > 0, \partial^2 F / \partial A^2 < 0$

We further assume that in addition to the timber products at the end of rotation, the forest area also yields some annual returns. These returns are from a number of non-timber forest products like fodder, fruits, leaves, soil and water conservation services, and other amenities as well as a wood product like fuelwood. Returns from these are usually available annually from the very first year. Let the per unit area yield of all such products which yield annual returns, be Q_{nt} units of some composite 'non-timber' forest product (NTFP), and let it fetch a price of P_{nt} per unit. The per unit area yield of this non-timber product is also assumed to be the same over different years for the period of consideration (this assumption is relaxed in the Appendix). We further suppose that the community gets all the available quantity of the NTFP, but its share as a fraction of final timber product at rotation is Z . We also assume that all returns are available at the end of the respective year.

Since most of the JFM agreements are for one rotation, let us first consider the case of one rotation only. In this situation, the owners' fixed threat point is $S_o = W_o R A$, and its present value is

$$PV(S_o) = W_o A [\exp(rR) - 1] / \exp(rR) [\exp(r) - 1]$$

and the community's fixed threat point is $S_c = W_c R A$ and its present value is

$$PV(S_c) = W_c A [\exp(rR) - 1] / \exp(rR) [\exp(r) - 1]$$

Then, the present value of the owner's pay-off gain is

$$Y_1 = \frac{(1-Z)PQ_t}{\exp(rR)} - \frac{W_o A [\exp(rR) - 1]}{\exp(rR) [\exp(r) - 1]}$$

and the present value of the community's pay-off gain is

$$Y_2 = \frac{ZPQ_t}{\exp(rR)} - \frac{(W_c - P_{nt} Q_{nt}) A [\exp(rR) - 1]}{\exp(rR) [\exp(r) - 1]}$$

The present value of the total pay-off gain $Y = Y_1 + Y_2$ is

$$Y = \frac{PQ_t}{\exp(rR)} - \frac{(W_o + W_c - P_{nt} Q_{nt}) A [\exp(rR) - 1]}{\exp(rR) [\exp(r) - 1]}$$

Our problem, now, is one of determining the optimal level of forest area (A), rotation (R) and share of community (Z) in the final timber produce. In light of the above situation, the maximum total pay-off gain with the option of negotiating the forest area (A) and rotation (R), is given by the first-order conditions

$$\partial Y / \partial A = 0 \text{ and } \partial Y / \partial R = 0$$

These two conditions, on simplification yield

$$P \frac{\partial Q_t}{\partial A} = \frac{(W_o + W_c - P_{nt} Q_{nt}) [\exp(rR) - 1]}{[\exp(r) - 1]}$$

and

$$P \frac{\partial Q_t}{\partial R} = rPQ_t + r \frac{(W_o + W_c - P_{nt} Q_{nt}) A}{[\exp(r) - 1]}$$

Suppose on solution these equations give $A = A_1$ and $R = R_1$ and for these values of A and R , $Q_t = Q_{t1}$.

Then the maximal total pay-off gain is

$$Y^* = \frac{PQ_{t1}}{\exp(rR1)} - \frac{(W_o + W_c - P_{nt}Q_{nt})A1[\exp(rR1) - 1]}{\exp(rR1)[\exp(r) - 1]}$$

The present values of the owner's and the community's pay-off gains for these optimal values of A and R can be represented by Y_1^* and Y_2^* , respectively.

The bargaining process will be complete, only when the share Z of the community is agreed upon by two parties. For distributional optimization, we find out the Nash unique solution which is given as

maximization of $L (= Y_1^* Y_2^*)$

The first-order condition for maximization is $dL/dZ = 0$ i.e.

$$\frac{d(Y_1^* Y_2^*)}{dZ} = 0$$

or

$$Y_1^* \frac{dY_2^*}{dZ} + Y_2^* \frac{dY_1^*}{dZ} = 0$$

Since

$$\frac{dY_1^*}{dZ} = -\frac{PQ_{t1}}{\exp(rR1)}$$

and

$$\frac{dY_2^*}{dZ} = \frac{PQ_{t1}}{\exp(rR1)}$$

On substituting these values the first-order condition becomes

$$Y_1^* = Y_2^*$$

Therefore, for maximal pay-off gains

$$\begin{aligned} & \frac{(1-Z)PQ_{t1}}{\exp(rR1)} - \frac{W_o A1[\exp(rR1) - 1]}{\exp(rR1)[\exp(r) - 1]} \\ &= \frac{ZPQ_{t1}}{\exp(rR1)} - \frac{(W_c - P_{nt}Q_{nt})A1[\exp(rR1) - 1]}{\exp(rR1)[\exp(r) - 1]} \end{aligned}$$

or

$$Z = 0.5 - \frac{(W_o - W_c + P_{nt}Q_{nt})A1[\exp(rR1) - 1]}{2PQ_{t1}[\exp(r) - 1]} \quad (1)$$

In case we wish to consider an infinite number of rotations, the basic formula for Z will remain same as in Eq. (1) above but $R1$, $A1$ and Q_{t1} will be replaced by respective optimum values of rotation (R), area (A) and timber quantity (Q_t) for the infinite rotations case^c.

The assumption made regarding negotiability of forest area and rotation may not be feasible in actual practice. For example, normally the community would have a clear idea of the forest area within the village boundary and would not like to give up any area included within it. Similarly, an economically impoverished community is likely to prefer early returns and need small timber like poles for its own consumption. The rotation therefore needs to be fixed according to the community's requirement. In such cases when rotation (R) and area (A) are not negotiable, the final forest produce quantity (Q_t) will be constant for a given area and rotation.

There may also be situations in which one of the parameters, either the rotation or the area, is negotiable while the other is not. For example, if there is a large forest area within the commu-

^cFor infinite rotations, given that $r > 0$

$$Y_1 = \frac{(1-Z)PQ_t}{\exp(rR) - 1} - \frac{W_o A}{\exp(r) - 1}$$

$$Y_2 = \frac{ZPQ_t}{\exp(rR) - 1} - \frac{(W_c - P_{nt}Q_{nt})A}{\exp(r) - 1}$$

$$Y = \frac{PQ_t}{\exp(rR) - 1} - \frac{(W_o + W_c - P_{nt}Q_{nt})A}{\exp(r) - 1}$$

Suppose on solution of first-order conditions for maximization of Y , we get $A = A2$, $R = R2$ and $Q_t = Q_{t2}$; then maximal pay-off gain is

$$Y^* = \frac{PQ_{t2}}{\exp(rR2) - 1} - \frac{(W_o + W_c - P_{nt}Q_{nt})A2}{\exp(r) - 1}$$

The Nash unique solution is given by $Y_1 = Y_2$, on substituting the values of Y_1 and Y_2 for optimal values of A and R we get

$$Z = 0.5 - \frac{(W_o - W_c + P_{nt}Q_{nt})A2[\exp(rR2) - 1]}{2PQ_{t2}[\exp(r) - 1]}$$

nity's land boundary and it feels that it cannot manage the total area, it may be ready to negotiate the extent of the area to be shared with the owner. Alternatively, there may be a situation in which there is no forest area within a village boundary but it is sufficiently close to invite use and the owner may be willing to hand over some of this area to the community. In this case Q_t will be a function of area (A) only. Similarly, there may be a situation in which there is only a small area (A) within a community's land boundary, and therefore it is fixed. However, the community may be economically well off, and may be able to wait for handsome returns in the future and so be ready to negotiate the rotation (R). In this case Q will be a function of the rotation length (R) only. By modifying and incorporating the actual nature and values of rotation (R), forest area (A) and quantity of final timber yield (Q_t), the situation-specific value for the community's share of the final timber yield can be obtained with the help of the general model.

6. Discussion

In a situation where no significant annual returns from non-timber forest products are perceived ($Q_{nt} \approx 0$) and there is very low rate of time preference ($r \approx 0$), the general result in Eq. (1) is reduced to

$$Z = 0.5 - \frac{(W_o - W_c)A1R1}{2PQ_{t1}}$$

The fraction Z will depend upon the difference between the owner's and the community's threat pay-offs; and the relative values of the difference of total threat pay-off ($= (W_o - W_c)A1R1$), from total area during rotation, with respect to return from timber products at rotation ($= PQ_{t1}$). The community's share of the final timber product will be half, less than half and more than half when $W_o = W_c$, $W_o > W_c$ and $W_o < W_c$, respectively. Normally in an area where the illicit use of the forests by the communities is high and law enforcement difficult, W_o is close to zero or even negative. In such circumstances joint management will be possible

only when the community's share of the final produce is more than half of the output.

In a situation where the annual returns from non-timber forest products are perceived to be significant but there is still a low rate of time preference, the value of Z will be lower than that in the situation discussed above. The decrease will depend upon the perception of the extent of the contribution made by NTFP. The value of Z will be half, more than half and less than half when $W_o + P_{nt}Q_{nt} = W_c$, $W_o + P_{nt}Q_{nt} < W_c$ and $W_o + P_{nt}Q_{nt} > W_c$, respectively.

In situations where the perception of no annual returns or annual returns from NTFP exist but the rate of time preference is positive, the factors affecting Z will be the same as in the first and second cases discussed above, respectively, plus the rate of time preference. Owing to time preference, all pay-offs will be replaced by present values of pay-offs.

These outcomes permit an examination of two cases of JFM in West Bengal, India where the state government fixed the same share for all communities without consideration of other factors involved.

6.1. A case study from West Bengal (India)

Vast areas in three districts, i.e. Bankura, Midnapore and Purulia of the Indian state of West Bengal had luxuriant forest of Sal (*Shorea robusta* Gaertn) until the 1930s. The area was exploited during the British regime through the Zamindars (landlords) at a rotation of 5–15 years for pole crops, and gradually the condition of the forests deteriorated as no attention was paid to regeneration. With Independence in 1947 and the enactment of the Estate Acquisition Act in 1953 all forest areas held by the erstwhile Zamindars were taken over by the Forest Department (FD) in 1955, in an effort to protect these forests. However, continued pressure from the increasing population, particularly for fuel and fodder as well as other means of livelihood such as sale of non-timber forest products, rendered the entire forest area of about 403 802 ha to degraded Sal bushes by the mid-1960s (Malhotra et al., 1991).

The conventional measures pursued by the FD were found to be grossly inadequate even to protect the bushy young coppice shoots of Sal. Thus, a constant tussle ensued between the staff guarding the forest and the people who were basically dependent on it for their subsistence needs. The above situation led to: (1) the FD not receiving any revenue from the forest lands and incurring huge expenditures on maintaining staff and infrastructure in their futile attempts at controlling illicit exploitation; (2) the local communities not only facing acute shortage of fuel, fodder and other non-timber forest products but also of drinking and irrigation water. A significant proportion of the population had to migrate seasonally to other areas in search of wage employment. It was increasingly realized by the FD that without the people's involvement, Sal forests would have no future (Palit, 1990).

In 1972, the Divisional Forest Officer at Midnapore, initiated a participatory model in a cluster of 11 villages in Arabari block to protect 1272 ha of degraded Sal coppice forest. The overwhelming success of the experiment encouraged the FD to extend the concept of JFM to other areas. The interests of user group communities are represented by the Forest Protection Committees (FPC) in JFM. As of July 1990, 1611 FPC had been formed, protecting 191 756 ha of primarily degraded Sal coppice forest lands. Thus, nearly 47% of the entire forest land and 77% of the entire Sal coppice area in the three southwestern districts of West Bengal are now being jointly managed by the FD and the local communities (Malhotra et al., 1991).

Below^d we work out the values of the community's share as per Nash outcome of final timber harvest for two FPC from West Bengal.

The Sitarampur FPC was formed in 1986. The committee has a 300 ha forest area and 99 families as its members. Before protection the committee members were getting returns from some NTFP, i.e. fuelwood, fruits and seeds, and Kendu

leaves (for rolling local cigarettes called bidis). In monetary terms it was approximately equal to Rs 312.00 (CAN\$13.56, at 1992 rate of CAN\$1.00 = 23.00 Indian Rupees) per year per hectare. The government received no returns. So the value of threat pay-offs of the government and the community were equal to zero and Rs 312 (CAN\$13.56), respectively. Even some negative value could be assigned to the government's pay-off, if the salaries of the FD personnel in the area were considered. After formation of FPC, the returns from NTFP have increased to approximately Rs 970 (CAN\$42.17). The returns from final forest harvest are estimated to be Rs 20 000 (CAN\$869.56) per hectare at a rotation of 12 years. On substituting these values in Eq. (1), the community's share of the final timber harvest (Z) for a zero rate of time preference is

$$Z = (1/2) - [(0 - 312 + 970) \times 12 \times 300] / (2 \times 20,000 \times 300) = 0.31$$

The Arabari Forest Protection Committee was formed in 1977. The committee has a 1272 ha forest area and 618 families as its members. The returns to the community before formation of the FPC were only Rs 110.00 (CAN\$4.78) per hectare per year and after the JFM programme these have increased to Rs 350 (CAN\$15.21) per hectare per year. The estimate of returns from the final forest harvest is the same as in the case of Sitarampur.

For a government pay-off equal to zero, the value of the community's share in the final timber produce (Z) for a zero rate of time preference turns out to be 0.43. Values of Z , for different rates of time preference are given in Table 1.

By an executive order dated 12 July 1989, the state government of West Bengal made communities partners in forest management, sharing output with the government. These arrangements give all the NTFP to communities and an arbitrarily determined 25% share of the final timber harvest. Our results for the two FPC discussed above indicate that in the case of Sitarampur, the share of 25%, as fixed by the government may be enough to ensure the community's long-term cooperation in forest management be-

^dSource of data: Malhotra and Poffenberger (1989). For calculating the returns before protection, it has been assumed that only half of the fuelwood, Kendu leaves, mushrooms, and one-quarter of medicinal plants and seeds were available.

Table 1
Community share for different rates of time preference

Rate of time preference (%)	Community share (Z)	
	Sitarampur	Arabari
8	0.18	0.38
6	0.22	0.40
4	0.25	0.41
2	0.28	0.42
0	0.31	0.43

cause the real rate of time preference of the village may not be realistically less than 4% year⁻¹. However, we believe that in the case of Arabari 25% is inadequate as the community's share resulting from a bargaining game. If the communities of Sitarampur and Arabari have the same rate of time preference, which is most likely, a share of no less than 41% for Arabari may be required to provide an equal incentive for long-term cooperation in forest management.

Though different shares for different communities may make sense from our theoretical point of view, there is a likelihood that the community receiving lower shares may perceive the arrangements as iniquitous. This could create conflict and negate the whole idea of forest management through cooperation. Therefore, to sustain the community's cooperation, the government may have to fix the same share for every community of final timber harvest and keep this figure at the average of the bargaining game outcome of all FPC. Now, to sustain the cooperation of the FPC which have the bargaining game outcome higher than the share fixed in this manner, the government should make an extra effort to increase the returns from NTFP to these communities, so that the equilibrium solution of the bargaining game can be obtained at lower values of community shares (Z). As already mentioned such an extra effort includes improvements in the collection and storage techniques for NTFP, installation of local level processing units and strengthening marketing facilities. Consideration of the economic costs of these efforts is beyond the scope of this paper but they could be relatively small in the marginal sense if seen as part of an overall

economic development effort for the region.

In summary, we have attempted a methodology for fixing the share of a community of forest produce, so that its long-term interest can be maintained in cooperating with the owner. We have not attempted to explain the outcomes on the basis of our model. The reason for this is that there are no outcomes yet in terms of sustained success or failure of JFM systems, based on the issue of community shares in forest produce. But, we are suggesting that to make JFM a permanent reality anywhere, the community share should be fixed very carefully and this model may be able to provide useful guidelines for doing so.

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Appendix

To be more realistic the model could have the following assumptions:

(1) different rates of time preference of owner (government) and community, being r_o and r_c , respectively;

(2) different fixed threat point for government and community for different years, being $W_o(t)$ and $W_c(t)$, respectively, which are functions of time;

(3) annual returns from non-timber forest products also being time variant, and so represented as $Q_{nt}(t)$;

(4) NTFP returns are received continuously instead of annually.

The present value of fixed threat pay-off of owner (government) now is

$$S_o = A \int_0^R W_o(t) \exp(-r_o t) dt$$

Similarly, the present value of the community's threat pay-off is

$$S_c = A \int_0^R W_c(t) \exp(-r_c t) dt$$

Then the present value of the owner's pay-off gain is

$$Y_1 = \frac{(1-Z)PQ_t}{\exp(r_o R)} - A \int_0^R W_o(t) \exp(-r_o t) dt$$

and the present value of the community's pay-off gain is

$$Y_2 = \frac{ZPQ_t}{\exp(r_c R)} - A \int_0^R [W_c(t) - P_{nt} Q_{nt}(t)] \exp(-r_c t) dt$$

making the total pay-off gain

$$Y = \frac{(1-Z)PQ_t}{\exp(r_o R)} + \frac{ZPQ_t}{\exp(r_c R)} - A \int_0^R [W_c(t) - P_{nt} Q_{nt}(t)] \exp(-r_c t) dt - A \int_0^R W_o(t) \exp(-r_o t) dt$$

Now suppose the first-order conditions dY/dA and dY/dR for a maximum of Y , give $A=A_2$ and $R=R_2$, and a corresponding value of $Q_t=Q_{t2}$. By substituting these values, we can calculate the maximal pay-off gain.

For distributional maximization, the Nash solution will give

$$Y_1 = Y_2$$

that is

$$\frac{(1-Z)PQ_{t2}}{\exp(r_o R_2)} - A \int_0^{R_2} W_o(t) \exp(-r_o t) dt$$

$$= \frac{ZPQ_{t2}}{\exp(r_c R_2)} - A \int_0^{R_2} [W_c(t) - P_{nt} Q_{nt}(t)] \exp(-r_c t) dt$$

This equality on simplification gives

$$Z = \frac{1}{1 + \exp(r_o - r_c)R_2} - \frac{A_2}{PQ_{t2} [\exp(r_o - r_c)R_2 + \exp(r_c - r_o)R_2]} \int_0^{R_2} [W_o(t) - W_c(t) + P_{nt} Q_{nt}(t)] \exp(-rt) dt$$

where

$$\text{integral} = \int_0^{R_2} [W_o(t) \exp(-r_o t) - [W_c(t) - P_{nt} Q_{nt}(t)] \exp(-r_c t)] dt$$

In the case of $r_o = r_c = r$ the equation of Z reduces to

$$Z = 0.5 - \frac{N_2}{2PQ_{t2}} \int_0^{R_2} [W_o(t) - W_c(t) + P_{nt} Q_{nt}(t)] \exp(-rt) dt$$

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